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Structure and Method for an MRAM Device with a Multi-Layer Top Electrode

Abstract

A semiconductor structure includes a bottom electrode, a first tunneling junction disposed over the bottom electrode, a first top electrode disposed over the first tunneling junction, a second tunneling junction disposed over the bottom electrode and spaced apart from the first tunneling junction, and a second top electrode disposed over the second tunneling junction.

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Background/Summary

PRIORITY DATA [0001] This is a continuation application of U.S. patent application Ser. No. 17/814,917 filed on Jul. 26, 2022, which is a divisional application of U.S. patent application Ser. No. 16/510,296 filed on Jul. 12, 2019, now issued as U.S. Pat. No. 11,563,167, which claims priority to U.S. Provisional Patent Application Ser. No. 62/736,623 filed on Sep. 26, 2018, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

[0003] One advancement in some IC design and fabrication has been the developing of non-volatile memory (NVM), and in particular to magnetic random-access memory (MRAM). MRAM offers comparable performance to volatile static random-access memory (SRAM) and comparable density with lower power consumption to volatile dynamic random access memory (DRAM). Compared to NVM Flash memory, MRAM may offer faster access times and suffer less degradation over time. An MRAM cell is formed by a magnetic tunneling junction (MTJ) comprising two ferromagnetic layers which are separated by a thin insulating barrier, and operates by tunneling of electrons between the two ferromagnetic layers through the insulating barrier. Scaling of MRAM cells in advanced technology nodes is limited by the resolution limit of both lithography and etching techniques. At the lower resolution limit, MTJ size variation between MRAM cells within an MRAM cell array can degrade memory performance. Although existing approaches in MRAM device formation have generally been adequate for their intended purposes, they have not been entirely satisfactory in all respects. Accordingly, there exists a need for improvements in this area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIGS. 1A and 1B illustrate perspective views of a magnetic tunneling junction (MTJ) within an MRAM cell.

[0006] FIG. 1C illustrates an MRAM cell array, in accordance with an embodiment.

[0007] FIGS. 2A and 2B show a flow chart of a method for forming a magnetic random-access memory (MRAM) cell array, according to aspects of the present disclosure.

[0008] FIGS. 3, 4, 5, 6, 7, 8, 9, 10, 13, 14, 15, 16, 17, 18, 19, 20A, 20B, and 21 illustrate cross-

sectional views of a semiconductor structure during a fabrication process according to the method of FIGS. 2A-2B, in accordance with some embodiments.

[0009] FIGS. 11 and 12 illustrate perspective views of a semiconductor structure during a fabrication process according to the method of FIGS. 2A-2B, in accordance with some embodiments.

DETAILED DESCRIPTION

[0010] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0011] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within $\pm 10\%$ of the number described, unless otherwise specified. For example, the term “about 5 nm” encompasses the dimension range from 4.5 nm to 5.5 nm.

[0012] The present disclosure is generally related to semiconductor devices and fabrication methods. Some aspects of the present disclosure relate to scaling techniques for magnetic random-access memory (MRAM) cells. More particularly, the present disclosure is related to providing a magnetic tunneling junction (MTJ) together with top and bottom electrodes and a method that achieves a substantially uniform pattern of MTJs within MRAM cells of an MRAM cell array, and further achieves a minimum MTJ dimension below the lower resolution limit of some optical lithography techniques.

[0013] In the forming of an MRAM cell, the critical dimension (CD) of an MTJ affects many operating parameters of integrated circuits, such as form factors, speed performance, power consumption, and thermal stability of an MRAM cell array. For an MRAM cell array patterned at the lower resolution limit of optical lithography, non-uniformity in the MTJ CD within the MRAM cells drives a non-uniform current density and variation in read/write characteristics between the MRAM cells. Non-uniformity in the dimensions of the MTJ also leads to a variation in coercivity between the MRAM cells due to the size variation. Additionally, the size and density of the MTJ are limited by the lower resolution limit of optical lithography (e.g., a lower limit of about 30 nm). Variation in the volume of an MTJ also leads to non-uniform thermal stability which also contributes to variable magnetoresistance between the MRAM cells. To achieve a minimum MTJ CD and to reduce MTJ CD bias, a multi-layer (e.g., a bilayer or a tri-layer) top electrode is formed above an MTJ stack, in some embodiments of the present disclosure. Each sub-layer of the multi-layer top electrode is constructed with different conductive material compositions, resulting in high etching selectivity of an upper sub-layer (e.g., the topmost sub-layer of a bilayer top electrode) to one or more bottom sub-layers during an etching process. Patterning of the top electrode in one etching process may cause damages to the underlying MTJ stack by the etchants applied. In

embodiments of the present disclosure, the high etching selectivity between the upper and bottom sub-layers allows the upper sub-layer and the bottom sub-layer(s) to be patterned separately. During the etching of the upper sub-layer, the bottom sub-layer(s) of the top electrode functions as an etching stop layer which protects the underlying MTJ stack from being damaged during the etching. The high etching selectivity also allows trimming the upper sub-layer with substantially vertical sidewalls without forming a footing profile, which would otherwise enlarge the MTJ CD. A patterning process then transfers a pattern in the upper sub-layer of the top electrode to the underlying sub-layers and the MTJ stack, while utilizing the upper sub-layer as a patterning mask. The present disclosure also helps to maintain the uniformity of the MTJ CD at a wafer scale.

[0014] FIGS. 1A and 1B illustrate perspective views of an MTJ **100** within an MRAM cell. The MTJ **100** includes an upper ferromagnetic plate **102** and a lower ferromagnetic plate **104**, which are separated by a thin insulating layer **106**, also referred to as a tunnel barrier layer. One of the two ferromagnetic plates (e.g., the lower ferromagnetic plate **104**) is a magnetic layer that is pinned to an antiferromagnetic layer, while the other ferromagnetic plate (e.g., the upper ferromagnetic plate **102**) is a “free” magnetic layer that can have its magnetic field changed to one of two or more values to store one of two or more corresponding data states.

[0015] The MTJ **100** uses tunnel magnetoresistance (TMR) to store magnetic fields on the upper and lower ferromagnetic plates **102** and **104**. For sufficiently thin insulating layer **106** thicknesses (e.g., about 10 nm or less), electrons can tunnel from the upper ferromagnetic plate **102** to the lower ferromagnetic plate **104**. Data may be written to the cell in a variety of ways. In one method, current is passed between the upper and lower ferromagnetic plates **102** and **104**, which induces a magnetic field stored in the free magnetic layer (e.g., the upper ferromagnetic plate **102**). In another method, spin-transfer-torque (STT) is utilized, wherein a spin-aligned or polarized electron flow is used to change the magnetic field within the free magnetic layer with respect to the pinned magnetic layer. Other methods to write data may be used. However, all data write methods include changing the magnetic field within the free magnetic layer with respect to the pinned magnetic layer.

[0016] The electrical resistance of the MTJ **100** changes in accordance with the magnetic fields stored in the upper and lower ferromagnetic plates **102** and **104**, due to the magnetic tunnel effect. For example, in FIG. 1A, the magnetic fields of the upper and lower ferromagnetic plates **102** and **104** are aligned (see arrows **112A** and **114A**), resulting in a low-resistance state (i.e., a logical “0” state). In FIG. 1B, a current has been passed through the MTJ **100** to induce a change in the magnetic field of the magnetic free layer (e.g., upper ferromagnetic plate **102**). Therefore, after this data write operation the magnetic fields in the upper and lower ferromagnetic plates **102** and **104** oppose one another (see arrows **112B** and **114B**), which gives rise to a high resistance state (i.e., a logical “1” state). Hence, by measuring the electrical resistance between the upper and lower ferromagnetic plates **102** and **104**, read circuitry coupled to the MTJ **100** can discern between “0” and “1” data states.

[0017] FIG. 1C illustrates an MRAM cell array **120**, which includes M rows (words) and N columns (bits) of MRAM cells. Each MRAM cell comprises an MTJ **100**. Word lines WL.sub.1, WL.sub.2, . . . WL.sub.M extend across respective rows of memory cells and bit lines BL.sub.1, BL.sub.2, . . . BL.sub.N extend along columns. To write data to a row of cells, a word line (WL) is asserted to select a row and appropriate biases are applied to the individual bit lines (BLs) to write respective values to the respective MRAM cells of the selected row. When data is written to or read from multiple cells along a row (e.g., a multi-bit word), data errors can occur for one or more MRAM cells due to manufacturing variations across the memory array that leads to erroneous bits being written to the MRAM cell array **120**. Therefore, there is a need to maintain CD uniformity at the lower resolution limit of manufacturing techniques.

[0018] FIGS. 2A and 2B illustrate a flow chart of a method **200** for forming an MRAM cell array in accordance with an embodiment. The method **200** is merely an example, not intended to limit the

present disclosure beyond what is explicitly recited in the claims. Additional operations can be provided before, during, and after the method **200**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the method. The method **200** is described below in conjunction with FIGS. **3-21**, which illustrate various cross-sectional and perspective views of an MRAM cell array during fabrication steps according to the method **200**. [0019] At operation **202**, the method **200** (FIG. **2A**) provides, or is provided with, a device structure **300** having a substrate **302**, such as shown in FIG. **3**. The substrate **302** is a silicon substrate in the illustrated embodiment. Alternatively, the substrate **302** may comprise another elementary semiconductor, such as germanium; a compound semiconductor including silicon carbide, gallium nitride, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and indium antimonide; an alloy semiconductor including silicon germanium, gallium arsenide phosphide, aluminum indium phosphide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and gallium indium arsenide phosphide; or combinations thereof. In another embodiment, the substrate **302** includes indium tin oxide (ITO) glass. In various embodiments, the substrate **302** is a wafer, such as a silicon wafer, and may include one or more epitaxially grown semiconductor layers in its upper portion.

[0020] The substrate **302** is disposed with an interconnect structure **304** on its upper surface. The interconnect structure **304** includes an inter-metal dielectric (IMD) layer **306** and a metal line **308** which extends horizontally through the IMD layer **306**. The IMD layer **306** can be an oxide, such as silicon dioxide, a low-k dielectric material such as carbon doped oxides, or an extreme low-k dielectric material such as porous carbon doped silicon dioxide. The metal line **308** can be made of a metal, such as aluminum, copper, or combinations thereof.

[0021] At operation **204**, the method **200** (FIG. **2A**) forms a dielectric-protection layer **310** over the interconnect structure **304**, such as shown in FIG. **4**. The dielectric-protection layer **310** is made of dielectric material, such as silicon oxide, silicon nitride, silicon oxynitride, or combinations thereof. The dielectric-protection layer **310** may be deposited through a process such as CVD, PVD or other suitable methods. In some embodiments, the dielectric-protection layer **310** comprises silicon carbide. In the illustrated embodiment, the dielectric-protection layer **310** comprises aluminum oxide having a thickness of approximately 200 Å. After the dielectric-protection layer **310** is formed, a mask **312**, such as a photoresist mask, is then formed over the dielectric-protection layer **310**. An etching process **314** is then carried out with the mask **312** in place to form patterned dielectric-protection layer **310**. The etching process **314** can be performed when a wet etchant or a plasma etchant is applied to the dielectric-protection layer **310** with the mask **312** in place, and forms an opening **316** which exposes a portion of the metal line **308**. The opening **316** can have a width w which corresponds to a width of a via in the interconnect structure **304**. The mask **312** can be removed after the etching.

[0022] At operation **206**, the method **200** (FIG. **2A**) forms a bottom electrode layer **320** over the dielectric-protection layer **310**, such as shown in FIG. **5**. The bottom electrode layer **320** extends downwardly into the opening **316** (FIG. **4**) to make direct electrical contact with the metal line **308**. The bottom electrode layer **320** may be a conductive material, such as, for example, titanium nitride, tantalum nitride, titanium, tantalum, or a combination of one or more of the foregoing. The bottom electrode layer **320** may be deposited through a process such as ALD, CVD, PVD or other suitable methods. For example, the bottom electrode layer **320** may be formed by depositing tantalum in a PVD process at a temperature above 300 degrees in Celsius. The bottom electrode layer **320** may be, for example, about 10-100 nanometers thick in some embodiments. The top surface of the bottom electrode layer **320** may have a non-planar top surface after deposition and may be planarized in one or more CMP processes thereafter.

[0023] Notably, the bottom electrode layer **320** is a continuous conductive body in direct electrical contact with the underlying metal line **308**, rather than relying on an extra via or contact feature for coupling. Because there is no extra via or contact feature between the bottom electrode layer **320**

and the underlying metal layer **308**, the overall height of the MRAM cell to be formed thereon can be reduced. This reduced overall height makes the MRAM cell more easily compatible with back-end-of-line (BEOL) process flows, providing better MRAM operations with reduced manufacturing cost.

[0024] At operation **208**, the method **200** (FIG. 2A) forms a magnetic tunneling junction (MTJ) stack **330** over an upper surface of the bottom electrode layer **320**, such as shown in FIG. 6. In the illustrated embodiment, the MTJ stack **330** includes a lower ferromagnetic electrode layer **332** (which can have a pinned magnetic orientation) and an upper ferromagnetic electrode layer **342** (which can have a free magnetic orientation). A tunneling barrier layer **340** is disposed between the lower and upper ferromagnetic electrodes layers **332** and **342**. The lower ferromagnetic electrode layer **332** can be a synthetic anti-ferromagnetic (SAF) structure that includes a bottom pinned ferromagnetic layer **334**, a top pinned ferromagnetic layer **338**, and a metal layer **336** sandwiched between the bottom and top pinned ferromagnetic layers **334** and **338**. Each layer of the MTJ stack **330** is disposed through appropriate layer growth techniques. Some layer growth techniques comprise sputter deposition, molecular beam epitaxy (MBE), pulsed laser deposition (PLD), atomic layer deposition (ALD) and electron beam (e-beam) epitaxy, chemical vapor deposition (CVD), or derivative CVD processes further comprising low pressure CVD (LPCVD), atomic layer CVD (ALCVD), ultrahigh vacuum CVD (UHVCVD), reduced pressure CVD (RPCVD), or any combinations thereof.

[0025] In some embodiments, the upper ferromagnetic electrode layer **342** comprises Fe, Co, Ni, FeCo, CoNi, CoFeB, FeB, FePt, FePd, or the like, and has a thickness ranging between approximately 8 Å and approximately 13 Å. In some embodiments, the tunneling barrier layer **340** provides electrical isolation between the upper ferromagnetic electrode layer **342** and the lower ferromagnetic electrode layer **332**, while still allowing electrons to tunnel through the tunneling barrier layer **340** under proper conditions. The tunneling barrier layer **340** may comprise, for example, magnesium oxide (MgO), aluminum oxide (e.g., Al.sub.2O.sub.3), NiO, GdO, Ta.sub.2O.sub.5, MoO.sub.2, TiO.sub.2, WO.sub.2, or the like. Further, the tunneling barrier layer **340** may be, for example, about 100-200 Å thick. In an embodiment, the bottom pinned ferromagnetic layer **334** includes a CoFeB film. Alternatively, the bottom pinned ferromagnetic layer **334** may comprise other materials, such as CoFeTa, NiFe, Co, CoFe, CoPt, CoPd, FePt, or an alloy of Ni, Co, and Fe. In an embodiment, the metal layer **336** includes ruthenium (Ru). Alternatively, the metal layer **336** may include other suitable material, such as Ti, Ta, Cu, or Ag. In some embodiments, the top pinned ferromagnetic layer **338** includes a ferromagnetic material substantially similar to that of the bottom pinned ferromagnetic layer **334**. For example, the top pinned ferromagnetic layer **338** includes a CoFeB film. In various embodiments, the MTJ stack **330** has a total height in a range from about 200 Å to about 400 Å.

[0026] At operation **210**, the method **200** (FIG. 2A) forms a multi-layer top electrode layer **350** over an upper surface of the MTJ stack **330**, such as shown in FIG. 7. In the illustrated embodiment, the multi-layer top electrode layer **350** is a bilayer that includes a lower electrode layer **352** and an upper electrode layer **354**. The lower and upper electrode layers **352** and **354** both comprise conductive materials, such as metal, metal nitride, metal carbide, or combinations thereof. The composition of the lower and upper electrode layers **352** and **354** are selected such that the upper electrode layer **354** has a high etch selectivity as compared to the lower electrode layer **352**. As to be shown in subsequent operations of the method **200**, the upper electrode layer **354** will be etched and trimmed to define a pattern to transfer to the MTJ stack **330**, during which the lower electrode layer **352** will function as an etch stop layer to protect the MTJ stack **330** from damage. In some embodiments, the upper electrode layer **354** comprises tantalum and the lower electrode layer **352** comprises titanium nitride. Alternatively, the lower electrode layer **352** may comprise other conductive materials, such as titanium, tungsten, tungsten nitride, or the like. In one specific example, the upper electrode layer **354** comprises tantalum and the lower electrode layer **352**

comprises tungsten carbide. The lower and upper electrode layers **352** and **354** may respectively be deposited through a process such as ALD, CVD, PVD, or other suitable methods. For example, the lower electrode layer **352** may be deposited in a PVD process at a temperature above 300 degrees in Celsius, and the upper electrode layer **354** may be subsequently deposited in a PVD process in-situ at a temperature above 300 degrees in Celsius.

[0027] The inventors have observed that when the thickness of the lower electrode layer **352** is larger than about 35% of the MTJ stack **330**, the MTJ stack **330** can be effectively protected from being oxidized during subsequent operations. Furthermore, when the thickness of the upper electrode layer **354** is larger than 110% of the MTJ stack **330** and the lower electrode layer **352** is larger than 35% of the MTJ stack **330**, the MTJ stack **330** can be protected from damages otherwise caused by fluorine or chlorine containing molecules that might penetrate into MTJ stack layers during subsequent operations. In the illustrated embodiment, the lower electrode layer **352** has a thickness in a range of about 35% to about 90% of the MTJ stack **330**, and the upper electrode layer **354** has a thickness in a range of about 110% to about 300% of the MTJ stack **330**. In various embodiments, the lower electrode layer **352** may have a thickness in a range from about 70 Å to about 180 Å, and the upper electrode layer **354** may have a thickness in a range from about 220 Å to about 500 Å.

[0028] In some embodiments, the lower electrode layer **352** may further include two or more sub-layers. Each sub-layer comprises conductive materials, such as metal, metal nitride, metal carbide, or combinations thereof. When the lower electrode layer **352** has two sub-layers, together with the upper electrode layer **354**, the top electrode layer **350** is regarded as a tri-layer stack. From top to bottom, in one example, the top electrode layer **350** comprises a Ta/Ti/W tri-layer stack. In another example, the top electrode layer **350** comprises a Ta/W/TiN tri-layer stack. In another example, the top electrode layer **350** comprises a Ta/TiN/WN tri-layer stack. In yet another example, the top electrode layer **350** comprises a Ta/WN/WC tri-layer stack.

[0029] At operation **212**, the method **200** (FIG. 2A) forms a hard mask layer **360** over the top electrode layer **350**, such as shown in FIG. 8. The hard mask layer **360** may include one or more patterning layers. In some embodiments, the hard mask layer **360** comprises dielectric material such as silicon oxide, silicon nitride, silicon oxynitride, amorphous carbon (APF), or suitable combination thereof. The hard mask layer **360** may be deposited through a process such as CVD or other suitable methods. The hard mask layer **360** may have a thickness ranging from about 150 Å to about 850 Å.

[0030] Operation **212** further includes patterning the hard mask layer **360** by photolithography and etching processes, such as shown in FIGS. 9 and 10. The photolithography and etching processes may first form and pattern a resist layer **370** over the hard mask layer **360**, then pattern the hard mask layer **360** using the patterned resist layer **370** as an etching mask. The resist layer **370** may include two or more layers, such as a bottom anti-reflective coating (BARC) layer **372** and a photosensitive layer **374** in the illustrated embodiment. An exemplary photolithography process may include a lithographic exposure to perform on the photosensitive layer **374** that exposes selected regions to radiation. The exposure causes a chemical reaction to occur in the exposed regions of the photoresist. After exposure, a developer is applied to the photoresist. The developer dissolves or otherwise removes either the exposed regions in the case of a positive resist development process or the unexposed regions in the case of a negative resist development process. Suitable positive developers include TMAH (tetramethyl ammonium hydroxide), KOH, and NaOH, and suitable negative developers include solvents such as n-butyl acetate, ethanol, hexane, benzene, and toluene. After the photoresist is developed, a pattern formed in the photosensitive layer **374** is transferred to the BARC layer **372** by selectively etching through openings in the developed photosensitive layer **374**, resulting in a patterned resist layer **370**. Subsequently, such as shown in FIG. 10, the exposed portions of the hard mask layer **360** may be removed by an etching process **376**, such as wet etching, dry etching, Reactive Ion Etching (RIE), ashing, and/or other

etching methods, resulting in a patterned hard mask layer **360**. The patterned hard mask layer **360** may be in a form of a pillar with a circular shape from a top view. The shape of the patterned hard mask layer **360** will be transferred to the top electrode layer **350** and the underlying MTJ stack **330** by patterning methods disclosed in subsequent embodiments herein. After patterning the hard mask layer **360**, the resist layer **370** may be removed.

[0031] Alternatively, operation **212** may use a polymer self-assembling process instead of a regular photolithography process to form a patterned dielectric layer over the hard mask layer **360**, such as a pillar array. An exemplary polymer self-assembling process includes first spin-coating a copolymer solution comprising first and second polymer species. The copolymer solution is subjected to self-assembly into a phase-separated material comprising a regular pattern of micro-domains of the second polymer species within a polymer matrix comprising the first polymer species. The first polymer species is then removed resulting with a pattern of micro-domains of the second polymer species. The pattern is then transferred to lower material layers by etching through while utilizing the pattern of micro-domains formed by the second polymer species as an etching mask.

[0032] In the illustrated embodiment, a block copolymer **380** is deposited by spin-coating the hard mask layer **360** with a copolymer solution comprising the first and second polymer species, such as shown in FIG. **11**. The copolymer solution may comprise poly(styrene-block-methylmethacrylate) (PS-b-PMMA), wherein the first polymer species comprises poly(methyl methacrylate) (PMMA), and the second polymer species comprises polystyrene (PS). In some embodiments, the volume fraction of the first and second polymer species is larger than about 0.6:0.4. The device **300** is then annealed through a thermal anneal or solvent anneal process (e.g., a UV irradiation), which results in a self-assembly of the copolymer solution into a phase-separated material (i.e., the block copolymer **380**). The anneal process promotes cross-linking of the PS molecules through the removal of one hydrogen from a benzene-bonded carbon of the PS polymer chain, such that two PS polymer units or chains of such ionized units may cross-link. The anneal process simultaneously degrades the PMMA polymer unit through the removal of one hydrogen from a methylenide molecule (CH₂) bonded to two carbons of the PMMA polymer unit. After the anneal process, the block copolymer **380** comprises a polymer matrix **384** of the first polymer species (PMMA), and a plurality of cylindrical micro-domains **382** of the second polymer species (PS), as shown in FIG. **11**. In some embodiments, the block copolymer **380** comprises a thickness of less than about 1,000 Å.

[0033] The first polymer species may then be removed (i.e., the polymer matrix **384**) through an oxygen reactive ion etch (RIE), which leaves the pattern of micro-domains **382** of the second polymer species, such as shown in FIG. **12**. In the illustrated embodiment, the micro-domains **382** comprise a pattern of polymer pillars residing in a periodic hexagonal close-packed (HCP) arrangement with minimum space(s) of less than about 30 nm. The micro-domains **382** then act as an etching mask to transfer the pattern to the hard mask layer **360**. In some embodiments, the micro-domains **382** are directly formed on the upper electrode layer **354** without forming the hard mask layer **360**, where the micro-domains **382** act as a patterned hard mask for subsequent patterning.

[0034] At operation **214**, the method **200** (FIG. **2B**) selectively etches the upper electrode layer **354** using the patterned hard mask layer **360** as an etching mask, such as shown in FIG. **13**. As discussed above, the compositions of the upper electrode layer **354** and the lower electrode layer **352** are selected such that the upper electrode layer **354** has a high etching selectivity as compared to the lower electrode layer **352**. As an example, operation **214** uses an etching process **388** that exhibits a larger than 10:1 etching selectivity of the upper electrode layer **354** over the lower electrode layer **352**. As a result, the etching process **388** removes portions of the upper electrode layer **354** not covered by the hard mask layer **360** and the lower electrode layer **352** remains relatively and/or substantially unchanged in thickness. Operation **214** may use a dry etching, a wet

etching, or other suitable etching processes. For example, a dry etching process may implement an oxygen-containing gas, a fluorine-containing gas (e.g., CF_4 , SF_6 , CH_2F_2 , CHF_3 , and/or C_2F_6), a chlorine-containing gas (e.g., Cl_2 , CHCl_3 , CCl_4 , and/or BCl_3), a bromine-containing gas (e.g., HBr and/or CH_3Br), an iodine-containing gas, other suitable gases and/or plasmas, and/or combinations thereof. For example, a wet etching process may comprise etching in diluted hydrofluoric acid (DHF); potassium hydroxide (KOH) solution; ammonia; a solution containing hydrofluoric acid (HF), nitric acid (HNO_3), and/or acetic acid (CH_3COOH); or other suitable wet etchant. In some embodiments, the etching process is an RIE process under process parameters such as a source power ranging from about 900 W to about 1200 W, a bias voltage ranging from about 0 V to about 500 V, a gas flow ranging from about 10 sccm to about 200 sccm, and a temperature ranging from about 15 degrees in Celsius to about 55 degrees in Celsius. The etchant gas may include Cl_2 , SiCl_4 , BCl_3 , NF_3 , N_2 , H_2 , CH_4 , HBr , He, Ar, or a combination thereof. The thickness of the lower electrode layer 352 is larger than about 35% of the MTJ stack 330, which effectively blocks etchant molecules from penetrating into the MTJ layers and from causing damages.

[0035] In some embodiments, operation 214 uses an end-point control method to precisely control the etching time. An end-point control analyzes emitted residuals in real time during an etching process, such as by inspecting light spectra emitted by a plasma during a plasma etching with an optical emission spectroscopy (OES). When material compositions from the lower electrode 352 start being detected, the etching process 388 stops at the upper surface of the lower electrode 352. In some embodiments, a pattern of cylindrical pillars in the patterned hard mask layer 360 is transferred to the upper electrode layer 354 when the etching process 388 stops. Each pillar may have a diameter ($D_{sub.1}$) about 40 nm.

[0036] At operation 216, the method 200 (FIG. 2B) trims the patterned upper electrode layer 354, such as shown in FIG. 14. The patterned upper electrode layer 354 is trimmed by a trimming process 390. In an embodiment, the trimming process 390 is a dry etching process. In a further embodiment, the trimming process 390 is an isotropic dry etching process. The etchant gas may include Cl_2 , SiCl_4 , BCl_3 , NF_3 , N_2 , H_2 , CH_4 , HBr , He, Ar, or a combination thereof. In some embodiments, the trimming process 390 may use another fluorine-containing gas (e.g., SF_6 , CH_2F_2 , CHF_3 , and/or C_2F_6), an oxygen-containing gas, a chlorine-containing gas (e.g., Cl_2 , CHCl_3 , CCl_4 , and/or BCl_3), a bromine-containing gas (e.g., HBr and/or CH_3Br), an iodine-containing gas, other suitable gases and/or plasmas, and/or combinations thereof. In various embodiments, the trimming process 390 uses an etchant that is selectively tuned to etch the patterned upper electrode layer 354 while the lower electrode layer 352 remains substantially unchanged. In one example, the trimming process 390 is an RIE process performed at a source power about 1500 W, a bias voltage ranging from about 0 V to about 50V, a gas flow ranging from about 10 sccm to about 250 sccm, a pressure ranging from about 5 to 40 mTorr.

[0037] The trimming process 390 helps to achieve a minimum critical dimension (CD) below the lower resolution limit of some optical lithography techniques. In an embodiment, the patterned upper electrode 354 is trimmed such that the trimmed diameter ($D_{sub.2}$) shrinks about 10% to 30% ($D_{sub.2}/D_{sub.1} \approx 0.7-0.9$). In the illustrated embodiment, $D_{sub.2}$ is about 30 nm, approximately 10 nm reductions from $D_{sub.1}$. It is noted that the trimming process 390 further removes any footing profile at the bottom of the patterned upper electrode layer 354, due to the highly directional isotropic dry etching. A footing profile may cause problems in CD bias when transferring a pattern to lower material layers. By eliminating footing profile, the trimming process further achieves a substantially straight sidewall that is perpendicular to the upper surface of the lower electrode layer 352. The vertical sidewall assists to transfer the vertical profile to MTJ stack 330 in subsequent operations. After trimming the patterned top electrode layer 354, the patterned

hard mask layer **360** may be removed.

[0038] At operation **218**, the method **200** (FIG. 2B) etches the lower electrode layer **352** and the MTJ stack **330** using the trimmed top electrode layer **354** as an etching mask, such as shown in FIG. 15. Operation **218** includes an etching process **392** to etch through the lower electrode layer **352** and the MTJ stack **330**. In an embodiment, the etching process **392** is an isotropic dry etching process. In some embodiments, the etching process **392** transfers a pattern of pillar array (e.g., cylindrical pattern) into the lower electrode layer **352** first, followed by the removal of portions of the MTJ stack **330** using the upper and lower electrode layers **354** and **352** together as an etching mask. In some embodiments, the etching process **392** etches the lower electrode layer **352** and the MTJ stack **330** together with no intermediate etching step.

[0039] Operation **218** may further etch into a top portion of the bottom electrode layer **320** using the patterned top electrode layer **350** and the MTJ stack **330** as an etching mask, such as shown in FIG. 16. In some embodiments, the etching of the top portion of the bottom electrode layer **320** comprises applying a wet etchant or a plasma etchant for a predetermined period of time. After the removal of the top portion, the bottom electrode layer **320** can be regarded as comprising a central bottom electrode portion **320a** which extends downwardly through the dielectric-protection layer **310** to contact the underlying metal line **308** and a peripheral bottom electrode portion **320b** which surrounds the central bottom electrode portion **320a**. The peripheral bottom electrode portion **320b** may extend to physically connect with other peripheral bottom electrode portions of adjacent MRAM cells, thereby forming a word line (e.g., word line WL; in FIG. 1C) in an MRAM cell array. A sidewall of the central bottom electrode portion **320a** may further have a footing profile interfacing with an upper surface of the peripheral bottom electrode portion **320b**. In the illustrated embodiment, the central bottom electrode portion **320a** has a top surface smaller in area than a bottom surface.

[0040] After patterning the top electrode layer **350**, the MTJ stack **330**, and the bottom electrode layer **320**, for the sake of simplicity of description, the patterned top electrode layer **350** is denoted as the top electrode **350**, the patterned MTJ stack **330** is denoted as the MTJ **330**, and the patterned bottom electrode layer **320** is denoted as the bottom electrode **320**. In some embodiments, the sidewall S.sub.354 of the upper electrode **354** is substantially perpendicular to the top surface of the substrate **302**, while the sidewall S.sub.352 of the lower electrode **352** is tapered away from the sidewall S.sub.354. For example, the sidewall S.sub.354 may form an angle from about 88° to about 90° with respect to the top surface of the substrate **302**, and the sidewall S.sub.352 may form an angle from about 85° to about 88° with respect to the top surface of the substrate **302**. With the tapered sidewall S.sub.352, the lower electrode **352** has a larger bottom surface area than its top surface area, while the top surface area equals a bottom surface area of the upper electrode **354**. In some embodiments, the sidewall S.sub.330 of the MTJ **330** is further tapered away from the sidewall S.sub.352. For example, the sidewall S.sub.330 may form an angle from about 80° to about 85° with respect to the top surface of the substrate **302**.

[0041] At operation **220**, the method **200** (FIG. 2B) forms a sidewall spacer **394** covering sidewalls of the top electrode **350**, the MTJ **330**, and the bottom electrode **320**, such as shown in FIG. 17. The sidewall spacer **394** may comprise a dielectric material, such as silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, other dielectric material, or combinations thereof, and may comprise one or multiple layers of material. The sidewall spacer **394** may be formed by depositing a spacer material as a blanket layer over the device **300**. Then the spacer material is etched by an anisotropic etching process. Portions of the spacer material on the sidewalls of the top electrode **350**, the MTJ **330**, and the bottom electrode **320** remain and become the sidewall spacer **394**. The sidewall spacer **394** also covers a portion of the upper surface of the peripheral bottom electrode portion.

[0042] At operation **222**, the method **200** (FIG. 2B) forms an inter-metal dielectric (IMD) layer **396** covering the sidewall spacer **394**, such as shown in FIG. 17. The IMD layer **396** may comprise

tetraethylphosphosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. The IMD layer **396** may be formed by PECVD, FCVD, or other suitable methods. In some embodiments, the IMD layer **396** is formed of a low-k dielectric layer or an extreme low-k dielectric layer, to a thickness of approximately 2500 Å. If an extreme low-k dielectric layer is used, a curing process may be followed after depositing the extreme low-k dielectric layer to increase its porosity, lower the k value, and improve the mechanical strengths. The operation **222** also performs one or more CMP processes to planarize the top surface of the device **300**.

[0043] At operation **224**, the method **200** (FIG. 2B) performs further steps to complete the fabrication of the device **300**. For example, the method **200** may form a via opening and a trench opening (not shown) in the IMD layer **396** and deposit conductive materials therein to provide interconnections, such as the top electrode via (TEVA) **402** that lands on the top electrode **350** and the metal trench **404**, such as shown in FIG. 18. The metal trench **404** is part of a metallic interconnection layer, such as a bit line (e.g., bit line BL.sub.1 in FIG. 1C) in an MRAM cell array. The TEVA **402** electrically connects an MRAM cell to the metallic interconnection layer. In some embodiments, the TEVA **402** and the metal trench **404** comprise tungsten or copper. The TEVA **402** and the metal trench **404** can be formed by a damascene or dual-damascene process.

[0044] An alternative embodiment of an MRAM cell is illustrated in FIG. 19. At operation **206**, the method **200** (FIG. 2A) may deposit the bottom electrode layer **320** as a conformal layer.

Consequently, the bottom electrode layer **320** comprises a central bottom electrode portion **320a**, a step portion **320c**, and a peripheral bottom electrode portion **320b**, such that an upper surface of the central bottom electrode portion **320a** is recessed relative to an upper surface of the peripheral bottom electrode portion **320b**. The central bottom electrode portion **320a** extends downwardly through the dielectric-protection layer **310** to make direct electrical contact with the underlying metal line **308**. A step portion **320c** extends upwardly from the central bottom electrode portion **320a** and along the sidewalls of the dielectric-protection layer **310** to couple the central bottom electrode portion **320a** to peripheral bottom electrode portion **320b**. The central bottom electrode portion **320a**, step portion **320c**, and peripheral bottom electrode portion **320b** can be a continuous, seamless body of conductive material. The upper surface of the central bottom electrode portion **320a** can be substantially and continuously planar between the step portion **320c**, and a lower surface of the MTJ **330** is disposed on the upper surface of the central bottom electrode portion **320a**. An MRAM cell deposited above a recessed bottom electrode has a reduced overall height, making the MRAM cell more easily compatible with BEOL process flows. In some embodiments, upper surfaces of one or more sub-layers of the lower ferromagnetic electrode layer **332** are lower than an upper surface of the peripheral bottom electrode portion **320b**. In one example, an upper surface of the bottom pinned ferromagnetic layer **334** is lower than an upper surface of the peripheral bottom electrode portion **320b** and an upper surface of the metal layer **336** is above the upper surface of the peripheral bottom electrode portion **320b**. In another example, an upper surface of the metal layer **336** is lower than an upper surface of the peripheral bottom electrode portion **320b** and an upper surface of the top pinned ferromagnetic layer **338** is above the upper surface of the peripheral bottom electrode portion **320b**. Sidewall spacers **394** extend continuously over upper surfaces of the peripheral bottom electrode portion **320b**, step portion **320c**, and optionally outer portions of the central bottom electrode portion **320a**, and extend upwardly along sidewalls of the MTJ **330** and the top electrode **350**.

[0045] FIG. 20A illustrates a cross-sectional view of the MRAM cell array in FIG. 1C along the A-A line. FIG. 20B illustrates a cross-sectional view of the MRAM cell array in FIG. 1C along the B-B line. In FIG. 20A, a plurality of MRAM cells are deposited sequentially along a word line (e.g., the word line WL.sub.1 in FIG. 1C). The bottom electrode layer **320** is a continuous conductive body extending beneath each MTJ of the plurality of MRAM cells along the word line. The bottom

electrode layer **320** is equivalently in parallel connection with the word line (e.g., metal lines **308** embedded in the interconnect structure **304**), which reduces conductive resistance of the word line. In FIG. **20B**, another plurality of MRAM cells are deposited sequentially along a bit line (e.g., the bit line BL.sub.1 in FIG. **1C**). The TEVA **402** and metal trench **404** form a continuous conductive body extending above the MRAM cells as a bit line. The bottom electrode layer **320** is discontinued between two adjacent MRAM cells to avoid shorting different word lines, which may be formed by selectively etching portions of the peripheral bottom electrode portion between adjacent MRAM cells along a bit line. The selective etching process may include wet etching, dry etching, and/or other suitable etching methods, through a patterned hard mask (not shown).

[0046] FIG. **21** illustrates a cross-section view of the MRAM cell array in FIG. **1C** along the A-A line in some alternative embodiments. Similar to the device **300** illustrated in FIG. **20A**, a plurality of MRAM cells are deposited sequentially along a word line (e.g., the word line WL.sub.1 in FIG. **1C**). The bottom electrode layer **320** is a continuous conductive body extending beneath each MTJ of the plurality of MRAM cells along the word line. Different from the device **300** illustrated in FIG. **20A**, the device **300'** in FIG. **21** has the metal line **308** underlying only the first MRAM cell along a word line. The bottom electrode layer **320** under the other MRAM cells subsequent to the first MRAM cell along a word line does not extend downwardly through the dielectric-protection layer **310**. The continuous bottom electrode layer **320** alone is configured to work as the word line for all the MRAM cells in the same word line. The device **300'** simplifies the interconnect structure **304** and reduces fabrication cost in avoiding forming extra openings in the dielectric-protection layer **310**.

[0047] Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to a semiconductor device and the formation thereof. For example, embodiments of the present disclosure provide an MRAM cell array achieving a minimum MTJ critical dimension. The present disclosure also helps to maintain the uniformity of the MTJ CD at a wafer scale. Furthermore, formation of this MRAM cell array can be easily integrated into existing semiconductor fabrication processes.

[0048] In one exemplary aspect, the present disclosure is directed to a magnetic memory device. The magnetic memory device includes a bottom electrode; a magnetic tunneling junction disposed over the bottom electrode; and a top electrode disposed directly over the magnetic tunneling junction, wherein the top electrode includes a first top electrode layer and a second top electrode layer above the first top electrode layer, and wherein the second top electrode layer is thicker than the first top electrode layer. In some embodiments, a sidewall of the first top electrode layer is tapered away from a sidewall of the second top electrode layer. In some embodiments, the first top electrode layer includes TiN and the second top electrode layer includes Ta. In some embodiments, the first top electrode layer includes metallic carbide and the second top electrode layer includes Ta. In some embodiments, the first top electrode layer is thinner than the magnetic tunneling junction and the second top electrode layer is thicker than the magnetic tunneling junction. In some embodiments, the first top electrode layer is thicker than about 35% of a thickness of the magnetic tunneling junction and the second top electrode layer is thicker than about 110% of the thickness of the magnetic tunneling junction. In some embodiments, the bottom electrode has a central bottom electrode portion surrounded by a peripheral bottom electrode portion, and wherein a top surface of the central bottom electrode portion is above a top surface of the peripheral bottom electrode portion. In some embodiments, the top surface of the central bottom electrode portion is smaller than a bottom surface of the central bottom electrode portion. In some embodiments, a top surface area of the first top electrode layer equals a bottom surface area of the second top electrode layer.

[0049] In another exemplary aspect, the present disclosure is directed to a method. The method includes forming a bottom electrode layer over a substrate; forming a magnetic tunneling junction stack over the bottom electrode; forming a top electrode layer over the magnetic tunneling junction stack, the top electrode including a first top electrode layer and a second top electrode layer over

the first top electrode layer, the first top electrode layer and the second top electrode layer having different metallic compositions; forming a patterned dielectric layer over the top electrode layer; etching the second top electrode layer with the patterned dielectric layer as an etching mask and the first top electrode layer as an etching stop layer; and patterning the first top electrode layer and the magnetic tunneling junction stack with the etched second top electrode layer as a patterning mask. In some embodiments, the method further includes after the etching of the second top electrode layer, trimming a width of the etched second top electrode layer. In some embodiments, the method further includes patterning a top portion of the bottom electrode layer, thereby forming a central bottom electrode portion surrounded by a peripheral bottom electrode portion. In some embodiments, the etching of the second top electrode layer is controlled by detecting material compositions emitted from the first top electrode layer. In some embodiments, the first top electrode layer is thinner than the magnetic tunneling junction stack and the second top electrode layer is thicker than the magnetic tunneling junction stack. In some embodiments, the first top electrode layer includes TiN and the second top electrode layer includes Ta. In some embodiments, during the etching of the second top electrode layer, an etching selectivity of the second top electrode layer over the first top electrode layer is larger than 10:1. In some embodiments, the forming of the patterned dielectric layer includes forming polymer pillars in a polymer matrix during a polymer self-assembling process.

[0050] In yet another exemplary aspect, the present disclosure is directed to a method of forming a magnetic random-access memory (MRAM). The method includes providing a semiconductor substrate; forming a bottom electrode layer over the semiconductor substrate; forming a magnetic tunneling junction stack over the bottom electrode layer, wherein the magnetic tunneling junction stack includes a lower ferromagnetic layer, a tunneling barrier layer over the lower ferromagnetic layer, and an upper ferromagnetic layer over the tunneling barrier layer; forming a first top electrode layer over the magnetic tunneling junction stack; forming a second top electrode layer over the first top electrode layer, wherein the first and second top electrode layers includes different conductive material compositions; forming a mask layer over the second top electrode layer; patterning the mask layer; etching the second top electrode layer, thereby transferring a pattern in the patterned mask layer to the second top electrode layer; trimming a width of the second top electrode layer; and etching the first top electrode layer and the magnetic tunneling junction stack with the trimmed second top electrode layer as an etching mask. In some embodiments, the second top electrode layer is thicker than the first top electrode layer. In some embodiments, the forming of the first top electrode layer and the forming of the second top electrode layer both include performing a PVD process with a temperature above 300 degrees in Celsius.

[0051] The foregoing outlines features of several embodiments so that those of ordinary skill in the art may better understand the aspects of the present disclosure. Those of ordinary skill in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those of ordinary skill in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a bottom electrode; a first tunneling junction disposed over the bottom electrode; a first top electrode disposed over the first tunneling junction; a second tunneling junction disposed over the bottom electrode and spaced apart from the first tunneling junction; and a second top electrode disposed over the second tunneling junction.
2. The semiconductor structure of claim 1, wherein the bottom electrode comprises a first portion

- disposed directly below the first tunneling junction, a second portion directly below the second tunneling junction, and a third portion connecting the first portion and the second portion, wherein at least one of the first portion and the second portion has a top surface higher than a top surface of the third portion, and a bottom surface lower than a bottom surface of the third portion.
- 3.** The semiconductor structure of claim 2, further comprising a spacer layer disposed on sidewalls of the first tunneling junction, the first top electrode, and the first portion, and on the top surface of the third portion.
- 4.** The semiconductor structure of claim 2, wherein the at least one of the first and the second portions comprises a top portion and a bottom portion below the top portion, wherein the bottom portion is wider than the top portion.
- 5.** The semiconductor structure of claim 2, further comprising a dielectric layer disposed between the first portion and the second portion.
- 6.** The semiconductor structure of claim 1, further comprising: a first metal line disposed over and electrically connected to the first top electrode; and a second metal line disposed over and electrically connected to the second top electrode, wherein the bottom electrode extends lengthwise in a first direction, and wherein the first metal line and the second metal line extend lengthwise in a second direction different from the first direction.
- 7.** The semiconductor structure of claim 6, wherein the bottom electrode is a first bottom electrode; and wherein the semiconductor structure further comprises: a second bottom electrode adjacent to the first bottom electrode, a third tunneling junction disposed over the second bottom electrode, and a third top electrode disposed over the second tunneling junction, wherein the first metal line is disposed over and electrically connected to the third top electrode.
- 8.** The semiconductor structure of claim 1, wherein the bottom electrode has a top surface between the first tunneling junction and the second tunneling junction, wherein the bottom electrode has a sidewall connected to the top surface and having a footing profile.
- 9.** The semiconductor structure of claim 1, further comprising a dielectric layer disposed below the first tunneling junction and the second tunneling junction, wherein the bottom electrode extends through the dielectric layer, and wherein the dielectric layer is directly below the first tunneling junction, the second tunneling junction, or both.
- 10.** A semiconductor structure, comprising: a first bottom electrode; a first tunneling junction disposed over the first bottom electrode; a first top electrode disposed over the first tunneling junction; a spacer surrounding the first tunneling junction and disposed on a sidewall of the first bottom electrode, wherein the first bottom electrode extends beyond an outer sidewall of the spacer; a second bottom electrode; a second tunneling junction disposed over the second bottom electrode; a second top electrode disposed over the second tunneling junction; and a first metal line disposed over and electrically connected to the first top electrode and the second top electrode.
- 11.** The semiconductor structure of claim 10, wherein the first metal line extends lengthwise in a first direction, wherein the first bottom electrode extends lengthwise in a second direction different from the first direction.
- 12.** The semiconductor structure of claim 10, wherein the first bottom electrode comprises a top portion, a middle portion below the top portion, and a bottom portion below the middle portion, wherein the spacer is disposed on a sidewall of the top portion and a top surface of the middle portion.
- 13.** The semiconductor structure of claim 12, wherein the bottom portion is wider than the top portion.
- 14.** The semiconductor structure of claim 10, wherein the sidewall of the first bottom electrode has a footing profile.
- 15.** The semiconductor structure of claim 10, further comprising: a third tunneling junction disposed over the first bottom electrode; and a third top electrode disposed over the third tunneling junction, wherein the first tunneling junction and the second tunneling junction are arranged along

a first direction, and the first tunneling junction and the third tunneling junction are arranged along a second direction different from the first direction.

16. The semiconductor structure of claim 15, further comprising a second metal line disposed over and electrically connected to the third top electrode, wherein the first metal line and the second metal line extend lengthwise in parallel.

17. The semiconductor structure of claim 15, further comprising: a second metal line disposed below and electrically connected to the first bottom electrode; and a dielectric layer around the first bottom electrode and disposed directly below the third tunneling junction.

18. A semiconductor structure, comprising: a bottom electrode; a tunneling junction disposed over the bottom electrode; and a top electrode disposed over the tunneling junction, wherein the bottom electrode includes a top portion, a middle portion disposed below the top portion, and a bottom portion disposed below the middle portion, wherein the middle portion extends laterally beyond a sidewall of the top portion, and wherein the sidewall of the top portion includes a footing profile intersecting with a top surface of the middle portion.

19. The semiconductor structure of claim 18, wherein the bottom portion is wider than the top portion.

20. The semiconductor structure of claim 18, further comprising a spacer disposed on the sidewall of the top portion and the top surface of the middle portion.
